Glomerular (Nephritic) — Lupus Nephritis

1. Lupus nephritis wolf den

WHAT YOU SEE

LUPUS NEPHRITIS WOLF DEN, 30% kidney at Dx

WHAT IT ENCODES

Lupus nephritis is immune-complex glomerulonephritis from SLE and is one of the most serious organ manifestations, present in roughly 30% of patients at diagnosis and developing in over half eventually. It spans the full clinical spectrum from asymptomatic urinary findings to nephrotic syndrome to crescentic RPGN. On IF it classically shows a 'full house' pattern (IgG, IgA, IgM, C3, and C1q) — the high-yield immunopathology cue.

2. Anti-dsDNA + low complement

WHAT YOU SEE

A wolf flanked by archers firing arrows into a kidney while a gauge needle drops past a downward C3/C4 arrow — the archers are complement being 'fixed' (consumed) onto the kidney, so the dial falls; anti-dsDNA rising as complement falls signals a flare.

WHAT IT ENCODES

Anti-dsDNA antibodies and LOW C3/C4 (consumed by immune-complex deposition via the classical pathway) both correlate with lupus nephritis activity and flares; rising anti-dsDNA with falling complement signals an impending or active flare. Anti-Smith is highly specific for SLE but does not track activity. This places lupus among the LOW-complement nephritic diseases (with PSGN, MPGN, endocarditis).

BOARD TRAP

Complement is normal in lupus nephritis — it is LOW (C3/C4 fall with active disease) and is used to monitor flares; normal complement would point to anti-GBM or pauci-immune disease

3. Classes transform — repeat biopsy

WHAT YOU SEE

CLASSES TRANSFORM — REPEAT BIOPSY gate

WHAT IT ENCODES

Renal biopsy is essential because it assigns the ISN/RPS class (I–VI), which directly guides therapy and prognosis — serologies and urinalysis alone cannot. Classes can transform over time (e.g., V can shift to IV), so a repeat biopsy is warranted when the clinical course changes or therapy fails. Biopsy also distinguishes active, treatable inflammation from irreversible chronic scarring.

4. Class I — minimal mesangial

WHAT YOU SEE

Class I — minimal mesangial, normal LM sleeping wolf

WHAT IT ENCODES

Class I (minimal mesangial) shows normal light microscopy with mesangial immune deposits visible only on IF/EM. It carries an excellent prognosis and generally needs no nephritis-directed immunosuppression. Treatment is driven by extrarenal lupus activity.

5. Class II — mesangial prolif

WHAT YOU SEE

Class II — mesangial proliferative cubs

WHAT IT ENCODES

Class II (mesangial proliferative) shows mesangial hypercellularity and matrix expansion with mesangial deposits. It is generally mild with a good prognosis and usually does not require aggressive immunosuppression unless proteinuria is significant. Watch for transformation to a proliferative class.

6. Class III — focal <50%

WHAT YOU SEE

Class III — focal <50% wolves

WHAT IT ENCODES

Class III is FOCAL proliferative lupus nephritis involving fewer than 50% of glomeruli. Like class IV it is an active, inflammatory (proliferative) lesion that REQUIRES immunosuppression — induction with steroids plus MMF or cyclophosphamide. The III/IV proliferative classes are the ones you treat aggressively.

7. Class IV — diffuse ≥50%

WHAT YOU SEE

Class IV — diffuse ≥50%, crescents/fibrinoid necrosis snarling wolf

WHAT IT ENCODES

Class IV is DIFFUSE proliferative lupus nephritis affecting 50% or more of glomeruli — the MOST COMMON and MOST SEVERE class, with the worst renal prognosis. It shows endocapillary proliferation, wire-loop lesions (subendothelial deposits), fibrinoid necrosis, and crescents, presenting with an active nephritic picture (hematuria, RBC casts, hypertension, rising creatinine) often with heavy proteinuria. It demands prompt aggressive induction immunosuppression.

BOARD TRAP

Class IV is the mildest form — it is the MOST severe and most common; do not undertreat it. The diffuse proliferative lesion carries the worst renal prognosis

8. Class V — membranous

WHAT YOU SEE

Class V — membranous wolf with collar

WHAT IT ENCODES

Class V is membranous lupus nephritis — diffuse GBM thickening with subepithelial deposits — presenting with NEPHROTIC-range proteinuria rather than a nephritic picture. It can coexist with proliferative classes (e.g., 'III+V' or 'IV+V'), in which case treat as proliferative. Pure class V is prone to thrombotic complications (nephrotic state) and is managed with RAAS blockade ± immunosuppression for heavy/persistent proteinuria.

9. Class VI — sclerotic

WHAT YOU SEE

Class VI — sclerotic >90%, petrified wolf gravestone

WHAT IT ENCODES

Class VI is advanced sclerosing lupus nephritis with more than 90% globally sclerotic glomeruli — irreversible chronic scarring representing burnt-out disease. Immunosuppression does NOT help because there is no active inflammation to treat; management shifts to CKD/ESKD care and transplant planning. This is why biopsy matters: you must not immunosuppress a scarred, inactive kidney.

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10. Induction: MMF or IV cyclophosphamide

WHAT YOU SEE

A medicine shelf showing IV-cyclophosphamide and IV-steroid bottles for INDUCTION beside an MMF jar for MAINTENANCE, with add-on belimumab/voclosporin vials — the shelf layout maps the proliferative-class treatment sequence.

WHAT IT ENCODES

For proliferative disease (class III/IV ± V), INDUCTION is high-dose corticosteroids (often IV methylprednisolone pulses then oral prednisone) plus either mycophenolate mofetil (MMF, ~2–3 g/day) OR IV cyclophosphamide (e.g., low-dose 'Euro-Lupus' 500 mg every 2 weeks x6, or high-dose monthly pulses). MAINTENANCE is typically MMF or azathioprine. Newer add-ons include belimumab and the calcineurin inhibitor voclosporin. MMF is preferred over cyclophosphamide in those wishing to preserve fertility.

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11. Hydroxychloroquine baseline

WHAT YOU SEE

A pill bottle labeled HYDROXYCHLOROQUINE with a 'baseline for all' sign — the bottle sits under every other drug to show it is the background agent given to every lupus patient.

WHAT IT ENCODES

Hydroxychloroquine is recommended for ALL lupus patients (including those with nephritis, unless contraindicated) because it reduces flares, lowers renal damage, improves survival, and is safe in pregnancy. Monitor for retinal toxicity with baseline and periodic ophthalmologic exams. It is a background/baseline agent layered under disease-specific immunosuppression.

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12. APL → microthromboses

WHAT YOU SEE

A dark blood-filled pit captioned 'worse despite anticoag' — the pooled clots are antiphospholipid microthrombi clogging glomerular capillaries, a separate thrombotic injury that keeps worsening even on blood thinners.

WHAT IT ENCODES

Coexisting antiphospholipid antibodies can cause a renal thrombotic microangiopathy (glomerular capillary microthrombi) that worsens renal outcomes independent of immune-complex inflammation and may progress despite anticoagulation. Suspect APS-related TMA when renal function declines out of proportion to the biopsy class or with systemic thrombosis. This is a distinct, immunosuppression-resistant mechanism of injury.

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13. Avoid prolonged childbearing risk

WHAT YOU SEE

Avoid prolonged in childbearing, renal vein thrombosis

WHAT IT ENCODES

Active lupus nephritis sharply increases pregnancy complications (preeclampsia, fetal loss, flares), so disease should be quiescent for ~6 months before conception. Cyclophosphamide and MMF are teratogenic and must be stopped; switch to pregnancy-compatible agents (hydroxychloroquine, azathioprine) and continue low-dose aspirin. Nephrotic-range proteinuria also raises renal vein thrombosis risk.

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14. Progression to ESKD

WHAT YOU SEE

20% → ESKD sign, nephrotic inactive

WHAT IT ENCODES

Despite therapy, a meaningful minority — roughly 10–20%, higher with diffuse proliferative class IV — progress to end-stage kidney disease. Poor prognostic factors include class IV histology, high chronicity index, delayed treatment, persistent proteinuria, hypertension, and Black race. Achieving early complete remission of proteinuria is the strongest predictor of long-term renal survival.
